5.4 Bahdanau attention

- we studied the machine translation problem in Section 4.6, where we designed an encoder-decoder architecture, based on two RNNs, for sequence to sequence learning
- specifically, the RNN encoder transforms a variable-length sequence into a fixed-shape context variable, then the RNN decoder generates the output (target) sequence token by token, based on the generated tokens and the context variable
- however, even though not all the input (source) tokens are useful for decoding a certain token, the *same* context variable that encodes the entire input sequence is still used at each decoding step

5.4 Bahdanau attention

- in a separate but related challenge of handwriting generation for a given text sequence, Graves designed a differentiable attention model to align text characters with the much longer pen trace, where the alignment moves only in one direction
- inspired by the idea of learning to align, Bahdanau et al. proposed a differentiable attention model without the severe unidirectional alignment limitation
- when predicting a token, if not all the input tokens are relevant, the model aligns (or attends) only to parts of the input sequence that are relevant to the current prediction
- this is achieved by treating the context variable as an output of attention pooling

5.4 Bahdanau attention

- when describing Bahdanau attention for the RNN encoder-decoder below, we will follow the same notation in Section 4.6
- the new attention-based model is the same as that in Section 4.6, except that the context variable $\boldsymbol{c}$ is replaced by $\boldsymbol{c}_{t'}$ at any decoding time step t'
- suppose that there are T tokens in the input sequence, then the context variable at the decoding time step t' is the output of attention pooling:

$$\boldsymbol{c}_{t'} = \sum_{t=1}^{T} \alpha(\boldsymbol{s}_{t'-1}, \boldsymbol{h}_t)\boldsymbol{h}_t,$$

  where the decoder hidden state $\boldsymbol{s}_{t'-1}$ at time step $t'-1$ is the query, and the encoder hidden states $\boldsymbol{h}_t$ are both the keys and values, and the attention weight α is computed as in (6), using the additive attention scoring function defined by (7)
- slightly different from the vanilla RNN encoder-decoder architecture, the same architecture with Bahdanau attention is depicted in Figure 5

5.4 Bahdanau attention

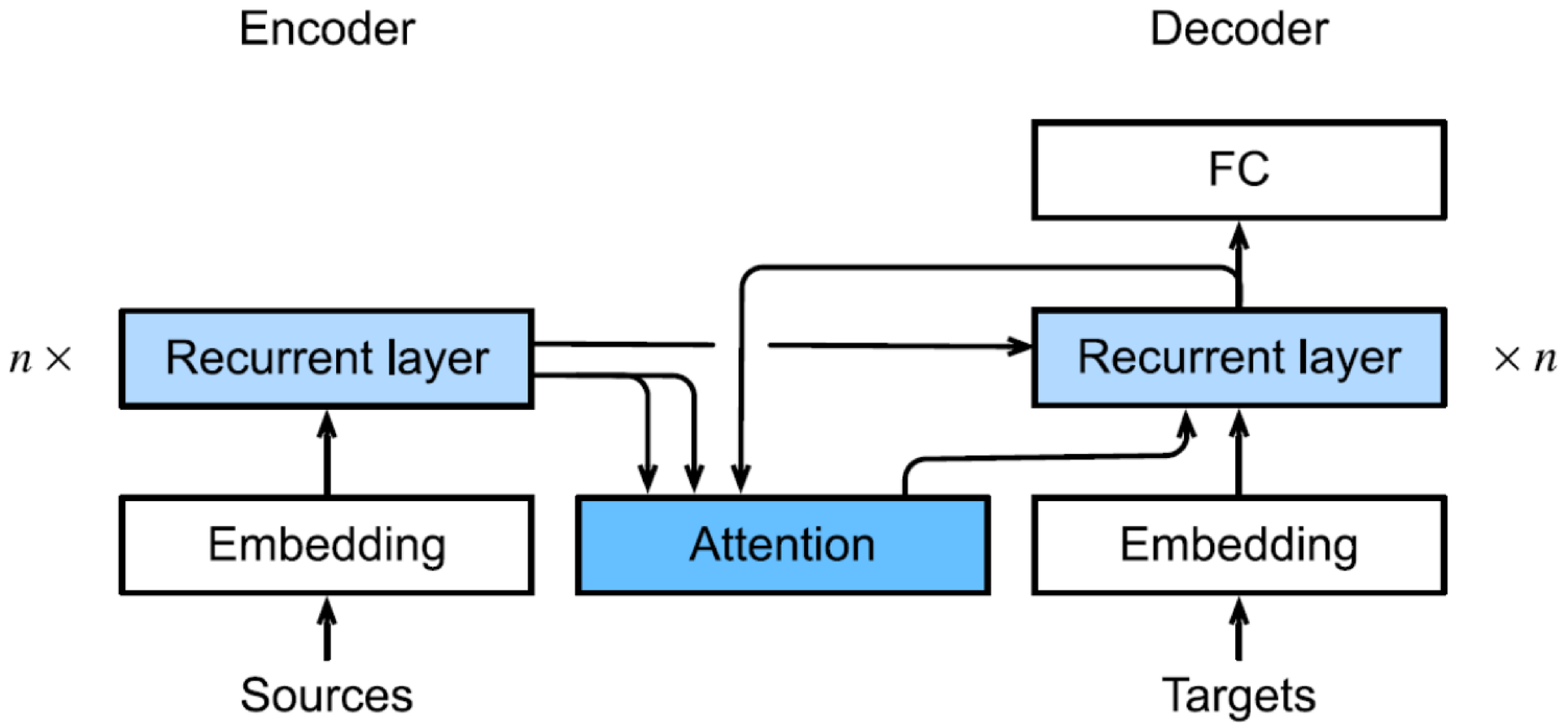


Figure 5: Layers in an RNN encoder-decoder model with Bahdanau attention.